Expected Points Added and Alliance Selection in the FIRST Robotics Competition

Abstract

Alliance captains in the FIRST Robotics Competition (FRC) select partners for the playoffs. This study examines how the top alliance's accepted first-round partner relates to candidates' Statbotics Expected Points Added (EPA) and qualification rank. Alliance records were linked to Statbotics data for 1,939 events from 2010 through 2025, excluding the canceled 2021 season. Conditional logit models compare accepted partners with other eligible teams at the same event. The primary measure is end-of-event EPA in raw game points. Robustness checks use qualification-end EPA, team-clustered confidence intervals, alternative rank forms, restricted choice sets, and a random-coefficient model. One additional EPA point was associated with higher selection odds (coefficient 0.213, 95% confidence interval [0.103, 0.323]); the qualification-end estimate was similar. A one-standard-deviation increase in EPA within an event had a coefficient of 2.490 [2.150, 2.829]. In the 2016–2025 component sample, autonomous and teleoperated EPA had confidence intervals above zero; the endgame interval included zero. Across four rank representations, adding rank to EPA increased McFadden pseudo-R^2 by 0.004–0.009, while adding EPA to rank increased it by 0.216–0.280. On the common 1,915-event sample, an EPA-plus-rank model identified 57.18% of accepted partners, compared with 34.62% for choosing the highest available qualification rank. Conclusions. Recorded selections align more closely with EPA than with qualification rank. The estimates do not establish whether captains used EPA or whether EPA-aligned choices improved playoff outcomes.

Keywords: FIRST Robotics Competition; alliance selection; Expected Points Added; performance measurement; discrete choice modeling; conditional logit

Introduction

The FIRST Robotics Competition (FRC) is an international high school robotics program. Students design, build, program, and operate a robot for a new game each season. Annual game manuals describe the rules, scoring system, and field tasks[9]. At official events, teams play qualification matches with changing partners and opponents. Match results and game-specific ranking points determine the standings. The highest-ranked teams then become alliance captains and select playoff partners.

Captains select partners for their scoring ability and their fit with the rest of the alliance. They may also consider reliability, defense, driver performance, and specialized tasks. The top-ranked captain makes the first invitation, before other captains have selected partners, so its choice has the clearest set of available candidates. This paper compares the top alliance's accepted first-round partner with the other available teams at each of 1,939 events from 2010 to 2025.

Public FRC statistics estimate team performance in different ways. Offensive Power Rating (OPR) uses alliance scores to estimate team-level scoring contributions[5]. Caleb Sykes later adapted Elo-style updating to FRC and published ratings and event tools that preceded Statbotics[11,35]. Statbotics' Expected Points Added (EPA) expresses a team's predicted average scoring contribution in game points, with autonomous, teleoperated, and endgame components[12,14]. Statbotics reports higher match-prediction accuracy for EPA than for its win-rate, OPR, and Elo baselines[14]. EPA does not directly measure defense, strategic compatibility, or breakdown risk, although captains may consider these when selecting a partner.

Qualification rank depends on match results and the ranking rules for that season. Scheduled partners, opponents, and bonus objectives can affect a team's standing. EPA estimates a team's contribution separately from the performance of its temporary alliances. Comparing the two measures shows how closely accepted partners align with estimated individual performance and with the official standings.

Studies of FIRST participants have examined goals, programming opportunities, self-efficacy, and postsecondary STEM engagement[26,27,31,32,39,40]. Research on robotics camps and competitions also reports changes in STEM knowledge, problem solving, attitudes, and career interest, with differences by gender and prior participation[3,20,28,29,34,38].

Other studies use FRC records to estimate competitive performance. Lim et al. estimate robot strengths and apply them to alliance-member selection[22]. Teng et al. study model selection and clusters of robot capabilities[36], and Galbiati et al. jointly model team formation and match winners[10]. These studies evaluate ratings or recommended teams. They do not test, across multiple seasons, which partner the top captain accepted.

Professional-sports drafts pose a related problem: teams choose candidates sequentially using incomplete measures of future performance. Draft studies examine uncertainty, institutional signals, tournament salience, rank categories, and decision biases[4,15–19,21,23,30,33]. These findings motivate testing several rank functions and comparing classification accuracy and incremental model fit, alongside changes in coefficients.

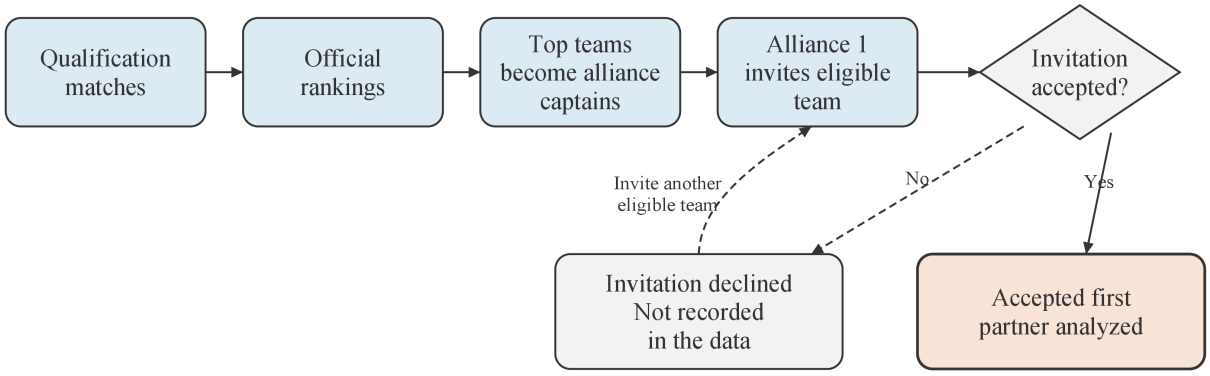
A conditional logit model relates differences in candidates' EPA and qualification rank to the partner accepted at each event[24,37]. The comparison is confined to teams facing the same game, field, and scoring rules. Reliability, compatibility, and other characteristics absent from the public data may also affect the choice.

The study tests three hypotheses. First, teams with higher EPA are more likely to become the accepted partner. Second, the autonomous, teleoperated, and endgame components have different associations with selection after accounting for their different scales. Third, EPA adds more information about selection than qualification rank does across several rank representations.

Figure 1 shows the alliance-selection process and the outcome examined here: the top alliance's accepted first-round partner. The analysis does not cover later picks, declined invitations, or playoff outcomes. The public data also omit private scouting information and team discussions, which limits what can be inferred about captains' reasons for choosing a partner.

The analysis links The Blue Alliance's alliance records with Statbotics data for 1,939 official events from 2010 to 2025. Pooled logistic regressions provide an initial comparison. The main conditional logit models compare candidates within events because raw EPA values have different

Figure 1. Procedure for Forming FRC Playoff Alliances



Solid path: recorded alliance formation. Dashed path: invitation history not observed.

Note: Most events use N=8 playoff alliances. The shaded outcome is the top alliance's accepted first-round partner. A declined invitation returns the choice to the captain. A declining team that is still in a captain position can remain or become a captain. When a captain accepts an invitation from a higher-ranked alliance, the next highest-ranked unassigned team becomes a captain. Some events have an additional backup or third-pick round. Public records do not consistently report invitation order.

meanings across games and scoring systems. The primary measure is end-of-event EPA. Repeating the model with qualification-end EPA checks whether the association remains when performance is measured before selection.

Higher event EPA is associated with a greater probability of becoming the accepted first-round partner. The qualification-end estimate is similar, suggesting that playoff updates do not drive the result. In the component sample, autonomous and teleoperated EPA have positive associations under team-clustered inference; the endgame interval includes zero. EPA adds more fit to rank-only models than rank adds to the EPA-only model.

Methods

Research Design

This observational study uses historical FRC records. Each observation represents a team at an event, so teams attending multiple events appear repeatedly. At each event, the analysis compares The Blue Alliance's recorded accepted first-round partner with the other eligible teams.

The top alliance chooses before other captains remove partners from the available set. Focusing on this choice makes the candidate set easier to identify. Each accepted partner is compared with candidates at the same event.

Sample

The sample covers official FRC events from 2010 to 2025, beginning with the first year of available event-level performance data. It excludes 2021, when the COVID-19 pandemic canceled the in-person competition season. Before matching alliance records, the team-level data contain 7,007 unique teams across 1,978 official events.

The estimation sample requires a recorded alliance selection and the relevant team performance data. It contains 1,939 events, 81,686 team–event observations, and 6,951 unique teams. The 39-event difference comprises 25 events without a valid top-alliance accepted partner, 13 Championship Finals fields where division winners advanced without a new selection, and one Festival of Champions event. The 1,978-event source file excludes historical offseason events because Statbotics does not provide EPA for them. Retained events are regionals, districts, district championships, district championship divisions, and championship divisions.

Data Collection

The data come from The Blue Alliance (TBA) and Statbotics[6,7,12,13]. TBA provides event identifiers, participating teams, qualification rankings, and playoff alliance members. Statbotics provides event EPA, qualification-end EPA, and EPA components. Matching event and team identifiers links each candidate's performance measures to the recorded alliance.

Each team–event record includes overall EPA and qualification rank. Autonomous, teleoperated, and endgame EPA are available from 2016, when Statbotics began reporting components. End-of-event EPA can include both qualification and playoff matches. Qualification-end EPA is recorded before alliance selection. TBA reports final alliance members but does not consistently report declined invitations.

Operational Definition of EPA

EPA is a sequential, Elo-style rating expressed in the current game's point units [12,14]. Its underlying rating is updated after match t according to the schematic rule

$$r_{i,t+1} = r_{i,t} + K_t\,(m_t - \widehat{m}_t) \tag{1}$$

where $m_t - \widehat{m}_t$ is the difference between the observed and predicted alliance scoring margin, and K_t controls the update size. Statbotics adjusts the update as matches accumulate and converts the underlying unitless rating to game points using the season's scoring distribution. The resulting EPA estimates the team's expected average point contribution under the model.

A team's rating entering an event includes information from its earlier matches and events that season. Event EPA therefore carries information beyond the current event. Statbotics updates the autonomous, teleoperated, and endgame components using game-specific score breakdowns. These components decompose scoring contribution, although the archived 2018 total and component means do not reconcile exactly. End-of-event EPA includes playoff updates; qualification-end EPA stops before selection and supplies the earlier measure for the timing check. The score data do not assign prevented points or mechanical reliability to individual robots, so EPA does not directly capture defense or breakdown risk.

The sample covers 15 competition seasons. Participation generally increased over this period, apart from 2020, when the season ended in early March. There was no in-person season in 2021. The 2025 sample includes 3,689 teams at 202 events. The main model compares candidates within each event, accounting for differences in the events represented over time.

Table 1 shows that regional events (791) and district events (893) account for most selections. The sample also includes 99 District Championships, 54 District Championship Divisions, and 102 Championship Divisions. Table 2 reports seasonal statistics. Mean EPA ranges from 1.2 points in 2010 to more than 130 points in 2018, reflecting differences in game scoring. These values are not directly comparable across seasons. The contribution of each match phase also varies: teleoperated EPA accounts for about three quarters of mean EPA in 2018 and one quarter in 2020. Pooling games without accounting for these scales can distort the component estimates.

Variables and Measurements

The outcome $\mathrm{D1}_{i,e}$ equals one when TBA records team i as the top-ranked alliance's accepted first-round partner at event e, and zero for the other candidates. The accepted partner may differ from the captain's first invitation. Without invitation-order or decline records, the analysis cannot

Table 1. Number of FRC events in the analysis sample by event type, 2010–2025

Event type	Number of events
Regional	791
District	893
District Championship	99
District Championship Div.	54
Championship Division	102
Total	1,939

Note: Counts are distinct FRC events by FIRST/The Blue Alliance event type for 2010–2025, excluding the canceled 2021 in-person season. The sample includes official events with Statbotics team-level EPA and a recorded alliance selection. Championship Finals (Einstein) fields are excluded because division-winning alliances advance intact without a new selection. Historical offseason events are excluded because Statbotics does not provide EPA for them.

Table 2. Descriptive statistics by season

Season	Events	Teams	Mean	SD	Auto	Teleop	Endgame
2010	49	1,611	1.2	1.0	–	–	–
2011	55	1,824	11.7	10.8	–	–	–
2012	68	2,207	7.6	5.9	–	–	–
2013	78	2,437	19.8	13.4	–	–	–
2014	102	2,694	31.2	18.5	–	–	–
2015	117	2,871	27.0	17.4	–	–	–
2016	134	3,113	25.5	9.2	7.1	14.3	4.1
2017	161	3,327	72.8	24.8	8.5	35.6	28.7
2018	176	3,613	132.5	28.4	17.6	100.0	15.5
2019	191	3,757	17.1	6.4	4.1	8.9	4.0
2020	52	1,956	27.2	14.1	9.3	6.3	11.5
2022	182	3,059	16.8	10.5	4.9	7.1	4.9
2023	183	3,288	28.8	13.8	7.9	14.4	6.3
2024	189	3,473	16.6	9.3	6.1	8.9	1.6
2025	202	3,689	30.4	18.0	6.6	20.3	3.5

Notes: Statistics use team–event observations in the estimation sample, excluding the top alliance's captain. Events and Teams count distinct events and teams within each season. Auto, Teleop, and Endgame are component means. EPA is expressed in each season's game points, so levels are not comparable across seasons. Components are available from 2016; dashes indicate missing component data. In 2018, the unrounded overall mean is 132.542 and the mean of the row-level component sums is 133.004. The -0.462 difference is present in the source snapshot. The 2020 season ended early, and the 2021 season was canceled.

separate the invitation from the acceptance. Declines may depend on rank: a team ranked 2 through 8 can decline and remain eligible to captain its own alliance.

The primary explanatory variable is end-of-event EPA in raw game points, referred to below as event EPA. A second specification uses qualification-end EPA to exclude matches played after selection. For seasons with component data, the analysis also uses autonomous, teleoperated, and endgame EPA. A secondary model standardizes EPA within each event as

$$z_{i,e} = \frac{\mathrm{EPA}_{i,e} - \overline{\mathrm{EPA}}_e}{s_{\mathrm{EPA},e}} \tag{2}$$

where $\overline{\mathrm{EPA}}_e$ and $s_{\mathrm{EPA},e}$ are the event mean and standard deviation. Subtracting the mean leaves conditional-logit probabilities unchanged because it cancels from the numerator and denominator. Dividing by $s_{\mathrm{EPA},e}$ changes the effective raw-point slope: a unit of $z_{i,e}$ represents a different number of EPA points at each event. The standardized specification allows the raw-point association to vary inversely with event dispersion. The raw-point model remains primary. The standardized model provides comparisons across games and event types, with sensitivity checks for the distribution of $s_{\mathrm{EPA},e}$ and the exclusion of low-dispersion events.

For the original rank specification, qualification rank is converted to

$$\mathrm{RankPct}_{i,e} = \frac{N_e - \mathrm{rank}_{i,e}}{N_e - 1} \tag{3}$$

where N_e is the number of teams at event e. The measure is 1 for the top-ranked team and 0 for the bottom-ranked team. Every event has more than one team, so the denominator is defined. To allow for a concentration of selections near the top of the standings, the analysis also uses $-\log(\mathrm{rank})$, 1/rank, and indicators for ranks 2 through 8, with rank 9 or worse as the reference group. Qualification rank is missing for 390 observations across 36 events. Every EPA-versus-rank comparison uses the same complete-case sample of 81,296 observations and 1,915 events.

Procedure

Alliance records are matched with EPA and qualification rankings to form an event-specific choice set. The top captain is removed because it cannot select itself. Pooled and conditional logit models compare the accepted partner with the remaining eligible teams.

Pooled models use all team–event observations and are also estimated by event type. Conditional models compare candidates within events. The analysis proceeds from overall EPA to its three components, then compares standardized EPA with four rank representations. Further checks exclude candidates ranked 2 through 8, reduce choice sets, estimate event-level random EPA slopes, examine individual seasons, and remove events with low EPA dispersion.

Data Analysis

The pooled logistic regression treats all team–event observations as one sample[1]. For event e and team i, it is

$$\Pr\left(\mathrm{D1}_{i,e} = 1\right) = \frac{\exp\left(\alpha + \beta \times \mathrm{EPA}_{i,e}\right)}{1 + \exp\left(\alpha + \beta \times \mathrm{EPA}_{i,e}\right)} \tag{4}$$

where $\mathrm{EPA}_{i,e}$ is team i's event EPA at event e. The coefficient β describes the association between EPA and selection. Its exponential, $\exp(\beta)$, is the multiplicative change in selection odds for a one-point increase in EPA.

The main conditional logit model represents the selection of one candidate from an event's choice set[24,37]. Let C_e contain all teams at event e except the captain. The probability that team i is selected is

$$\Pr\left(\text{team } i \text{ picked at event } e\right) = \frac{\exp\left(\beta \times \mathrm{EPA}_{i,e}\right)}{\sum_{j \in C_e} \exp\left(\beta \times \mathrm{EPA}_{j,e}\right)} \tag{5}$$

Selection probabilities depend on EPA differences within events. Event-level intercepts cancel from the comparison. Equation (5) corresponds to a logit model with event-specific fixed effects[2]. Standard errors in the pooled regressions are clustered by event because candidates' outcomes are mechanically linked within each choice set[8]. Confidence intervals for the principal conditional models are clustered by team to account for repeated appearances of 6,951 teams. Two-way clustering by event and team gave the same substantive conclusions.

The primary raw-EPA model is repeated with qualification-end EPA. Component models use raw points, within-event standard deviations, and within-event interquartile ranges. For each rank function, EPA-only, rank-only, and joint models are fitted on the same sample. The main comparison is the change in pseudo-R^2 when either EPA or rank is added. A restricted analysis removes candidates ranked 2 through 8 and retains only events where the accepted partner was ranked 9 or worse.

Conditional logit assumes independence of irrelevant alternatives (IIA): the odds of choosing one candidate over another do not depend on the other available teams. Choice-set checks retain only the top-ranked half of candidates or apply three seeded 25% random reductions. A Hausman–McFadden comparison tests coefficient stability. Because strict stability is rejected, the analysis also estimates a mixed logit with event-level random EPA slopes using 20-point Gauss–Hermite quadrature. This model allows the EPA slope to vary across events and relaxes the marginal IIA restriction of the fixed-slope conditional logit[37].

Model fit is summarized using McFadden's pseudo-R^2[25]:

$$R^2_{\mathrm{McF}} = 1 - \frac{\ell\left(\widehat{\beta}\right)}{\ell(0)} \tag{6}$$

where $\ell(\widehat{\beta})$ is the fitted model's log likelihood and $\ell(0)$ is the log likelihood when every eligible team has an equal selection probability within its event. A higher value indicates a larger improvement over equal-probability selection. McFadden's pseudo-R^2 does not measure the proportion of variation explained, as ordinary R^2 does in linear regression.

Ethical Considerations

The analysis uses publicly reported team-level competition data. It includes no student records, surveys, interviews, or information identifying individual students.

Data and Code Availability

The accompanying reproducibility package contains the public-source data snapshot, sample-construction scripts, model code, and exhibit generator. The documented pipeline rebuilds the estimation sample, robustness outputs, tables, and figures.

Results

Overall EPA and Alliance Selection

The pooled logistic regressions show a positive, statistically significant association between event EPA and selection across all major event types (Table 3). Coefficients range from about 0.009 to 0.015.

Pooled pseudo-R^2 values range from 0.03 to 0.05, indicating modest improvement over models without EPA. These models apply a common raw-point coefficient across seasons with different scoring scales. They provide a descriptive comparison; the within-event models are the main tests of the hypotheses.

An EPA of 30 may indicate a dominant robot in one season and an average robot in another. Pooling those values mixes scoring scales. The conditional logit compares candidates facing the same scoring system at the same event.

Within-Event Choice Model

The primary within-event model has an all-event coefficient of 0.213 [0.103, 0.323] (Table 4). One additional EPA point multiplies selection odds by about 1.24, or roughly 24%, among candidates at the same event. This is an increase in odds, not in percentage points of probability. The qualification-end coefficient is similar at 0.224 [0.108, 0.341], suggesting that playoff updates do not drive the association.

A one-standard-deviation increase in EPA within an event has a coefficient of 2.490 [2.150, 2.829]. The standardized model fits better while allowing the effective raw-point slope to vary

Table 3. Logit estimates of the association between EPA and the probability of being the top alliance's first pick, by event type

Event type	$\widehat{\beta}_{\text{EPA}}$	McFadden R^2	N	Events
Regional	0.0149(0.0007)	0.049	37,044	791
District	0.0125(0.0005)	0.039	30,584	893
District Championship	0.0097(0.0014)	0.026	4,155	99
District Championship Div.	0.0094(0.0013)	0.028	2,203	54
Championship Division	0.0116(0.0010)	0.041	7,700	102
All events combined	0.0122(0.0004)	0.038	81,686	1,939

Notes: Each row reports a separate descriptive logistic regression of the accepted-partner indicator on event EPA for the full 2010–2025 sample. Standard errors, in parentheses, are clustered by event because outcomes are linked within each choice set. The conditional models use team-clustered confidence intervals to account for repeated team appearances.

Table 4. Conditional-logit estimates for the top alliance's accepted first partner

Panel A: End-of-event and qualification-end EPA

Event type	End EPA $\widehat{\beta}$ [95% CI]	End R^2	Qual.-end EPA $\widehat{\beta}$ [95% CI]	Qual.-end R^2	N	Events
Regional	0.220 [0.062, 0.379]	0.557	0.230 [0.069, 0.392]	0.551	37,044	791
District	0.227 [0.067, 0.388]	0.548	0.236 [0.067, 0.404]	0.544	30,584	893
District Championship	0.149 [-0.165, 0.463]	0.358	0.197 [-0.270, 0.664]	0.410	4,155	99
District Championship Div.	0.187 [-0.424, 0.799]	0.443	0.187 [-0.415, 0.790]	0.445	2,203	54
Championship Division	0.170 [-0.140, 0.480]	0.456	0.166 [-0.138, 0.470]	0.450	7,700	102
All events combined	0.213 [0.103, 0.323]	0.531	0.224 [0.108, 0.341]	0.530	81,686	1,939

Panel B: Within-event standardized EPA and choice-set size

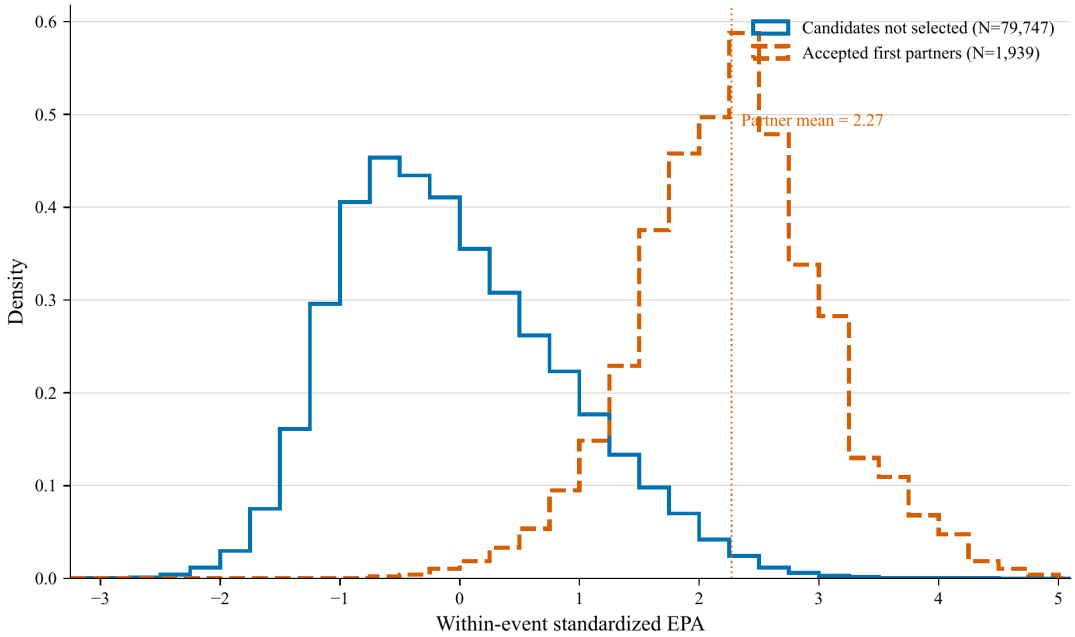
Event type	EPA z $\widehat{\beta}$ [95% CI]	McFadden R^2	Mean candidates	Median candidates	N	Events
Regional	2.368 [1.901, 2.835]	0.634	46.8	47	37,044	791
District	2.681 [2.123, 3.238]	0.622	34.2	36	30,584	893
District Championship	2.082 [1.375, 2.788]	0.465	42.0	44	4,155	99
District Championship Div.	2.365 [0.729, 4.002]	0.520	40.8	39	2,203	54
Championship Division	2.615 [1.308, 3.922]	0.535	75.5	74	7,700	102
All events combined	2.490 [2.150, 2.829]	0.610	42.1	39	81,686	1,939

Note: Each row is a separate conditional logit grouped by event, with team-clustered confidence intervals. Panel A reports McFadden pseudo-R^2 for end-of-event and qualification-end EPA. Panel B reports choice-set sizes alongside event-type fit statistics. Within-event standardization changes the effective raw-point slope according to event EPA dispersion. District Championship Divisions begin in 2017 and include 54 events. Their end-of-event and qualification-end coefficients are 0.187109 and 0.187101 before rounding; only 9 of 2,203 EPA pairs are exactly equal.

with event EPA dispersion. Panel B reports the mean and median number of candidates per event. These choice-set sizes differ by event type, so the event-type pseudo-R^2 values are not directly comparable measures of competitive similarity.

A separate audit checked the District Championship Division estimates. All 54 events occur from 2017 onward, and all 2,203 observations have component and rank data. Only 9 end-of-event EPA values exactly equal their qualification-end values. The coefficients, 0.187109 and 0.187101, round to the same value at three decimals despite the different underlying data.

Figure 2. Within-Event Standardized EPA for Selected and Nonselected Candidates



Note: The selected series has one accepted partner from each of 1,939 events. The nonselected series contains the other 79,747 candidates.

Accepted partners tend to have higher standardized EPA than nonselected candidates (Figure 2). On the common rank sample, choosing the highest-EPA candidate matches 55.25% of accepted partners. Defense, reliability, strategic fit, and declined invitations could help explain the remaining choices, but the public data do not distinguish their contributions.

EPA Components and Selection

Component data are available for 2016–2025, with 58,994 observations at 1,470 events. Table 5 reports raw-point estimates and estimates scaled by each component's within-event standard deviation and interquartile range.

The full-sample raw autonomous and teleoperated coefficients are 0.275 [0.157, 0.393] and 0.258 [0.051, 0.465]. Both intervals exclude zero, but their difference, 0.017 [-0.138, 0.173], is not statistically distinguishable from zero. The endgame estimate is 0.101 [-0.021, 0.222], with an interval that includes zero. Many event-type intervals also include zero. The estimates do not

Table 5. EPA components and the accepted first partner, 2016–2025

Panel A: Raw-point conditional-logit estimates by event type

Event type	Auto $\widehat{\beta}$ [95% CI]	Teleop $\widehat{\beta}$ [95% CI]	Endgame $\widehat{\beta}$ [95% CI]	McFadden R^2	N	Events
Regional	0.263 [0.096, 0.429]	0.262 [-0.067, 0.591]	0.101 [-0.085, 0.287]	0.590	22,675	501
District	0.315 [0.123, 0.506]	0.282 [-0.007, 0.571]	0.127 [-0.061, 0.315]	0.593	25,625	757
District Championship	0.168 [-0.104, 0.440]	0.185 [-0.391, 0.761]	0.022 [-0.273, 0.318]	0.370	3,266	84
District Championship Div.	0.229 [-0.284, 0.741]	0.205 [-0.537, 0.947]	0.039 [-0.838, 0.916]	0.461	2,203	54
Championship Division	0.230 [-0.130, 0.590]	0.215 [-0.372, 0.802]	0.035 [-0.371, 0.440]	0.483	5,225	74
All events combined	0.275 [0.157, 0.393]	0.258 [0.051, 0.465]	0.101 [-0.021, 0.222]	0.564	58,994	1,470

Panel B: Full-sample component effects on comparable scales

Scale of change	Auto $\widehat{\beta}$ [95% CI]	Teleop $\widehat{\beta}$ [95% CI]	Endgame $\widehat{\beta}$ [95% CI]	McFadden R^2	Events
Raw EPA points	0.275 [0.157, 0.393]	0.258 [0.051, 0.465]	0.101 [-0.021, 0.222]	0.564	1,470
One within-event SD	0.671 [0.537, 0.806]	1.851 [1.551, 2.150]	0.521 [0.397, 0.646]	0.619	1,470
One within-event IQR	0.790 [0.606, 0.974]	2.297 [1.854, 2.741]	0.539 [0.406, 0.672]	0.607	1,470

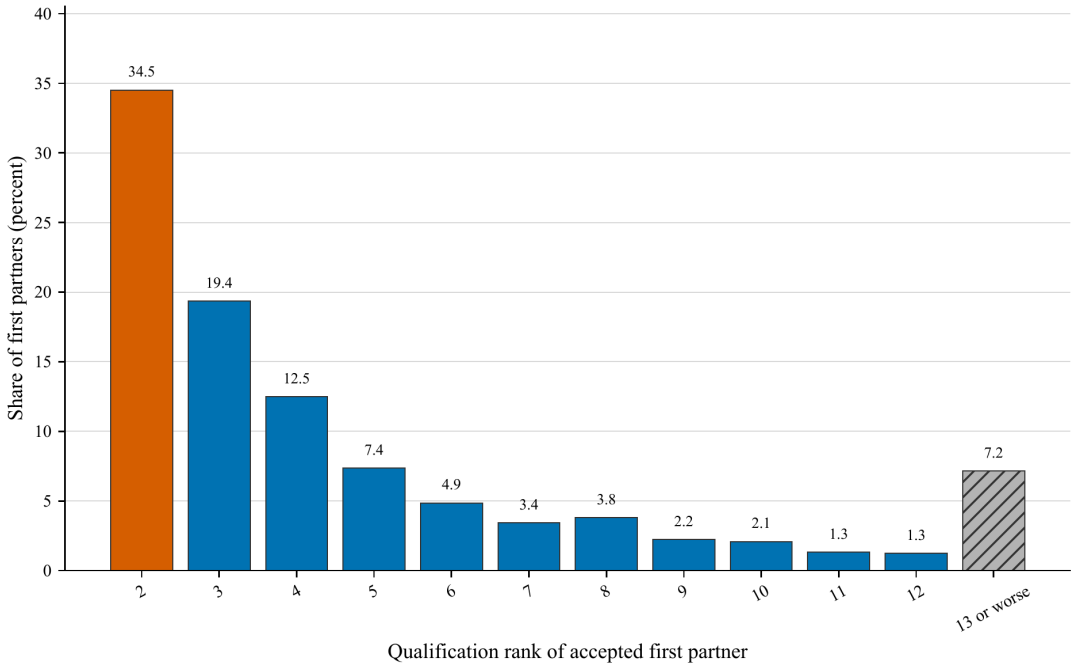
Note: Each row reports three coefficients from one conditional logit, with team-clustered confidence intervals. Component data begin in 2016. In Panel A, an interval containing zero indicates that the coefficient is not statistically distinguishable from zero at the 5 percent level. The full-sample auto and teleop intervals exclude zero; the endgame interval includes zero. Event-type estimates are less precise. Panel B scales each component by its within-event standard deviation and interquartile range to compare associations on common scales.

establish a positive association for every component within every event type.

The components have different raw-point variability. After within-event standardization, teleoperated EPA has the largest association, followed by autonomous and endgame EPA. Supplementary Table A1 examines why the pooled teleoperated coefficient changes sign. It rises from −0.023 to 0.086 after adding season indicators and to 0.954 after within-season standardization. The components remain substantially correlated. Differences in seasonal scoring scales account for part of the sign reversal; collinearity and other specification differences may also contribute.

EPA versus the Official Ranking

Figure 3. Qualification Rank of the Top Alliance's Accepted First Partner



Note: Rank 1 is the top captain and is excluded from the candidate set. The bar for ranks 13 or worse pools several ranks, so it is not directly comparable with individual-rank bars.

Figure 3 shows that rank 2 is the accepted partner in 34.52% of the 1,915 complete-rank events and rank 3 in 19.37%.

Table 6 compares model fit on the same 81,296 observations. Rank enters as a linear percentile, negative log rank, inverse rank, or indicators for ranks 2 through 8. EPA-only pseudo-R^2 is 0.616; rank-only values range from 0.340 to 0.408, and joint values from 0.620 to 0.624. Adding rank to EPA raises pseudo-R^2 by 0.004–0.009. Adding EPA to rank raises it by 0.216–0.280. EPA contributes more incremental fit under all four rank specifications.

Removing candidates ranked 2 through 8 leaves 271 events whose accepted partner was ranked 9 or worse. In this subset, adding rank raises pseudo-R^2 by 0.011, and adding EPA raises it by 0.339. The relative fit gains remain in the same order. This restricted sample cannot recover invitation sequences or establish how declines bias the EPA-versus-rank comparison.

Table 6. EPA versus alternative representations of qualification rank

Rank form	EPA-only R^2	Rank-only R^2	Joint R^2	Add rank ΔR^2	Add EPA ΔR^2	Joint EPA $\widehat{\beta}$ [95% CI]	Top-1 (%)
Linear rank percentile	0.616	0.408	0.624	+0.009	+0.216	2.215 [2.088, 2.342]	57.2
-log(rank)	0.616	0.404	0.622	+0.007	+0.218	2.253 [1.664, 2.843]	57.5
1/rank	0.616	0.340	0.620	+0.004	+0.280	2.329 [2.000, 2.659]	57.2
Indicators for ranks 2–8	0.616	0.378	0.622	+0.006	+0.243	2.297 [1.961, 2.634]	56.9

Note: All models use the same 81,296 team–event observations at 1,915 events with complete EPA and rank data. R^2 is McFadden's pseudo-R^2 relative to equal-probability selection. Confidence intervals are clustered by team. Changes in fit are the primary comparison. The indicator specification includes separate terms for ranks 2 through 8, with rank 9 or worse as the reference category. EPA adds more fit to each rank-only model than rank adds to EPA.

Table 7. Prediction benchmarks on the common EPA-and-rank sample

Rule or model	Correct picks	Events	Accuracy (%)
Uniform random choice (event-specific)	49.1	1,915	2.57
Choose highest-EPA candidate	1058	1,915	55.25
Choose highest available qualification rank	663	1,915	34.62
Always choose qualification rank 2	661	1,915	34.52
EPA-only conditional logit	1058	1,915	55.25
EPA plus linear-rank conditional logit	1095	1,915	57.18

Note: All benchmarks use the same 1,915 events. The random benchmark is calculated within each event and then averaged. With a positive EPA-only coefficient, the model's top-1 prediction is the highest-EPA candidate. The highest-available-rank rule uses the ranks observed in the data; the next row always selects rank 2.

On the common sample in Table 7, event-weighted random choice matches 2.57% of accepted partners. Choosing the highest available qualification rank matches 34.62%, choosing rank 2 matches 34.52%, choosing the highest EPA matches 55.25%, and the EPA-plus-rank model matches 57.18%. These in-sample rates describe agreement with accepted partners. They do not measure playoff performance.

Season, Dispersion, and Choice-Set Robustness

The standardized-EPA coefficient is positive in every observed season, with wider intervals in the shortened 2020 season (Supplementary Figure A1). Supplementary Figure A2 shows the event-level dispersion used to standardize EPA. Excluding events with EPA standard deviations below 1, or the bottom 1% or 5% of the dispersion distribution, gives coefficients of 2.515, 2.509, and 2.532, respectively. These estimates are close to the full-sample value of 2.490, suggesting that low-dispersion events do not drive the result.

Restricting choice sets gives mixed evidence about IIA. Keeping the top-ranked half changes the standardized-EPA coefficient from 2.531 to 2.421; three seeded random reductions give estimates from 2.490 to 2.553. The Hausman–McFadden test rejects strict coefficient stability under the top-half restriction. A random-slope mixed logit gives a positive mean EPA coefficient of 2.608 [2.469, 2.748] and a modestly lower AIC than the fixed-slope model. The EPA association persists across these specifications, although the rejected stability test limits the fixed-slope interpretation.

Discussion

Interpretation

Higher-EPA teams were more likely to become the top alliance's accepted first-round partner, supporting the first hypothesis. The primary raw-point coefficient is 0.213 [0.103, 0.323], equivalent to about 24% higher selection odds per EPA point. The similar qualification-end estimate, positive seasonal estimates, and low-dispersion checks suggest that this pattern is not confined to one timing measure or subset of events.

The component results give partial support to the second hypothesis. Autonomous and teleoperated raw-point coefficients are not statistically distinguishable from each other, although both contrasts with endgame are positive. The full-sample endgame interval includes zero under team-clustered inference. Teleoperated EPA has the largest association on standardized scales. Raw coefficients therefore should not be read as fixed measures of each component's influence. Captains may value endgame ability for complementary alliance roles, but their preferences are not observed.

EPA adds more fit than rank under all four rank functions, supporting the third hypothesis. It raises rank-only pseudo-R^2 by 0.216–0.280, compared with gains of 0.004–0.009 when rank is added to EPA. Removing candidates ranked 2 through 8 leaves the same ordering, but confines the comparison to events with lower-ranked accepted partners. Rank contributes some additional

information. The comparison cannot establish whether captains consulted EPA.

The EPA-only and joint EPA-and-rank models match 55.25% and 57.18% of accepted partners, respectively. This measures how well the models reproduce recorded choices. Assessing those choices would require comparing playoff outcomes for EPA-aligned and other selections. Without that comparison, these results do not justify a rule for constructing pick lists.

The pooled teleoperated coefficient is negative without season controls and positive after adding season indicators or standardizing within seasons. Correlations among components can widen confidence intervals, but they do not by themselves explain this sign reversal. Component estimates pooled across FRC games need to account for changes in scoring scales.

Limitations

The public data omit scouting notes, reliability, defense, driver performance, compatibility, prior partnerships, and geographic relationships between captains and candidates. The archived estimation file also lacks rookie status and team age. If these characteristics correlate with EPA or rank, the coefficients may reflect some of their association with selection. EPA is based on scoring records and does not directly measure defensive contribution or breakdown risk.

Invitation order and declines are incompletely recorded. Teams ranked 2 through 8 may decline and remain captains, making this omission particularly relevant to the rank comparison. An accepted partner reflects both an invitation and an acceptance, which the data cannot separate. The direction of bias for EPA relative to rank is unknown. Restricting candidates by rank cannot recover these missing decisions.

The fixed-slope conditional logit assumes IIA. Although EPA remains positively associated with selection after random choice-set reductions and in the random-slope model, the top-half Hausman–McFadden test rejects strict coefficient stability. The fixed-slope estimates and fit statistics depend on that assumption. The analysis covers only the top alliance's first accepted partner; later rounds may favor different roles. End-of-event EPA includes playoff updates. Qualification-end EPA addresses this timing issue for overall EPA, but either measure may miss damage, late improvements, or strategic changes that captains observe at the event.

Future Work

Recording the full invitation sequence across every round of the serpentine draft would separate invitations from acceptances. It would also allow analysis of how the value of scoring, defense, reliability, and complementary roles changes as teams leave the available set.

Standardized scouting records could add observed team attributes and prior partnerships to the choice model. Linking accepted partners to playoff outcomes would allow a separate test of whether EPA-aligned choices perform better. Such an outcome comparison is needed before recommending EPA-based pick lists.

Conclusions

Across 1,939 official FRC events, higher EPA is associated with becoming the top alliance's accepted first partner. The association persists with qualification-end EPA and across the season, dispersion, choice-set, and random-slope checks. EPA adds more model fit than qualification rank under all four rank representations. Autonomous and teleoperated associations are positive, while the team-clustered endgame interval includes zero. Missing scouting information, unobserved declines, and the rejected IIA stability test limit what these patterns reveal about captains' decisions and the quality of their choices.

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